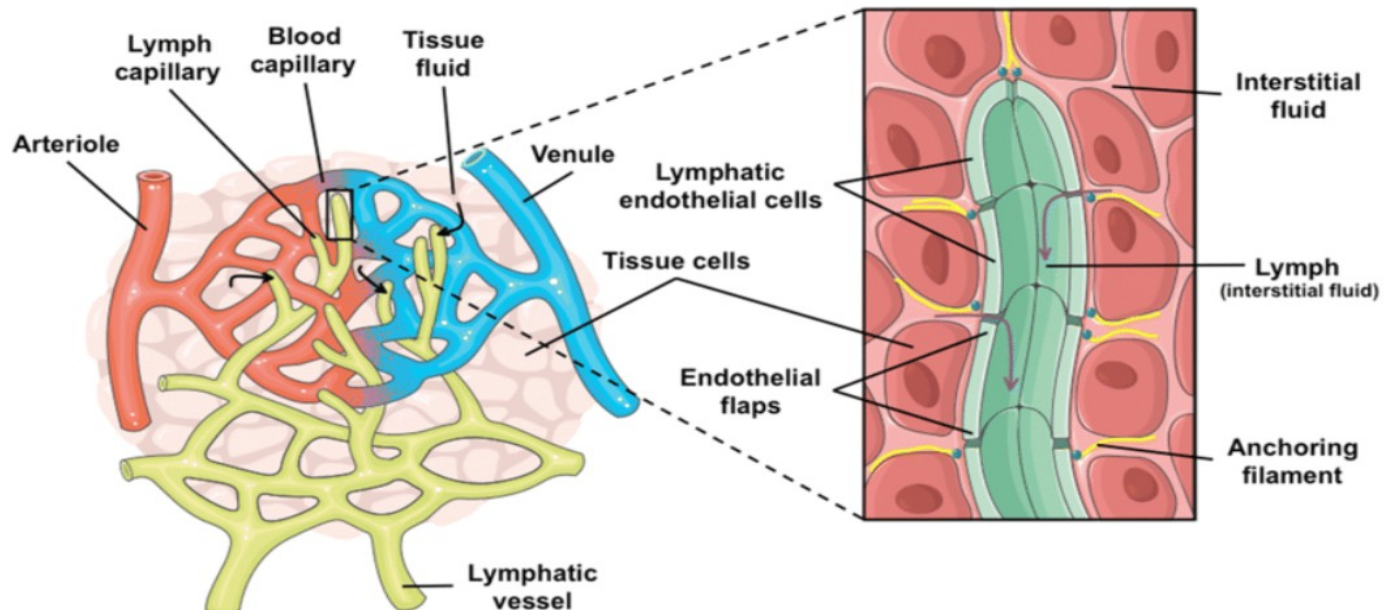


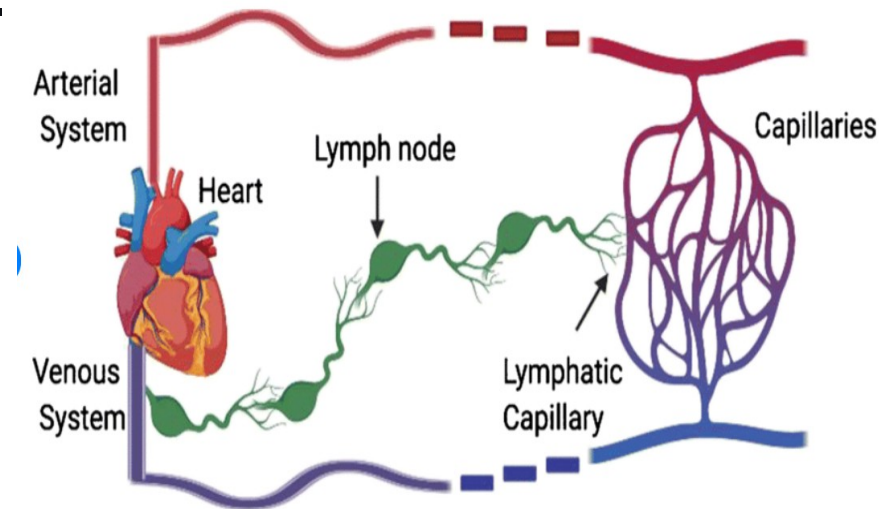
PATHOLOGY OF BLOOD VESSELS AND LYMPHATICS



BY MELESE A.(MD, PATHOLOGIST)

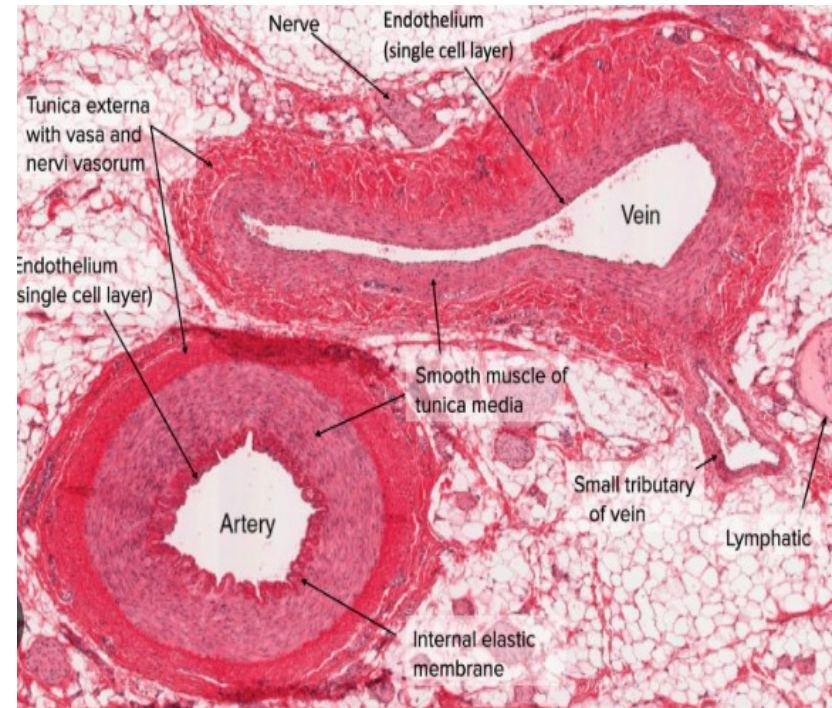
Introduction

- The circulatory system (blood and lymphatics) is **not merely** a set of passive pipes.
- It is a highly **dynamic, metabolically active** organ system lined by endothelial cells
- Regulate
 - ☐ vascular tone
 - ☐ Coagulation
 - ☐ inflammation
 - ☐ permeability
- Pathology arises when these regulatory functions are disrupted.



Normal Structure & Function

- Blood vessels have a layered (tunica) structure:
 - **Tunica Intima:** Inner lining of endothelial cells resting on a basement membrane and a thin layer of connective tissue.
 - **Tunica Media:** Middle layer of smooth muscle cells and elastic fibers (varies by vessel type). Responsible for vasoconstriction and vasodilation.
 - **Tunica Adventitia:** Outer layer of connective tissue containing nerves, lymphatics, and **vasa vasorum** (small vessels that supply the walls of larger vessels).



Responses to Injury

- The endothelial cell is the key responder.
- Injury (hemodynamic, biochemical, or immunological) leads to:
 - ❑ **Endothelial Dysfunction:** Increased permeability, upregulation of adhesion molecules, and a shift toward a pro-coagulant, pro-inflammatory state.
 - ❑ **Smooth Muscle Cell Activation:** Migration from media to intima, proliferation, and production of extracellular matrix, a central event in many vascular diseases.

Major Pathologic Processes

A. Arteriosclerosis: "Hardening of the Arteries"

1. Atherosclerosis

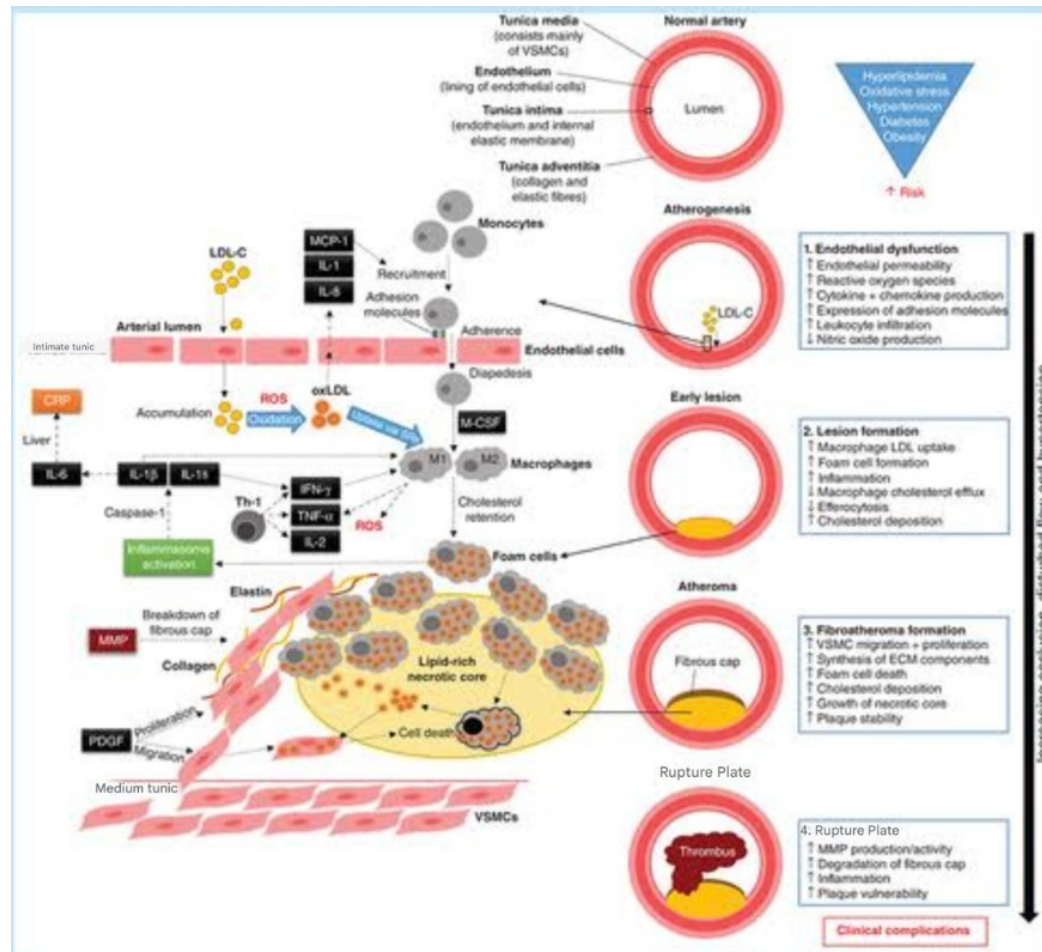
- **The most common and clinically significant form**

1. Pathogenesis:

- A chronic inflammatory response to endothelial injury in large and medium-sized arteries.

2. Key Steps:

- Endothelial dysfunction → Accumulation of LDL cholesterol in intima → Oxidation of LDL → Monocyte recruitment (become macrophages) → Macrophages engulf ox-LDL to become **foam cells** (earliest lesion, **fatty streak**) → Release of growth factors → Smooth muscle cell migration/proliferation → Formation of a fibrous cap over a necrotic lipid core (**atheromatous plaque**).



Atherosclerosis

3. Complications:

- 1. Rupture/Erosion:** Exposes thrombogenic material, leading to thrombosis and acute occlusion (e.g., myocardial infarction, stroke).
- 2. Hemorrhage into Plaque:** Expands the plaque rapidly.
- 3. Aneurysm:** Weakening of the medial wall leading to dilation.
- 4. Calcification:** "Hardening" of the plaque.

4. Clinical Effects: Coronary artery disease, cerebrovascular disease, peripheral arterial disease.

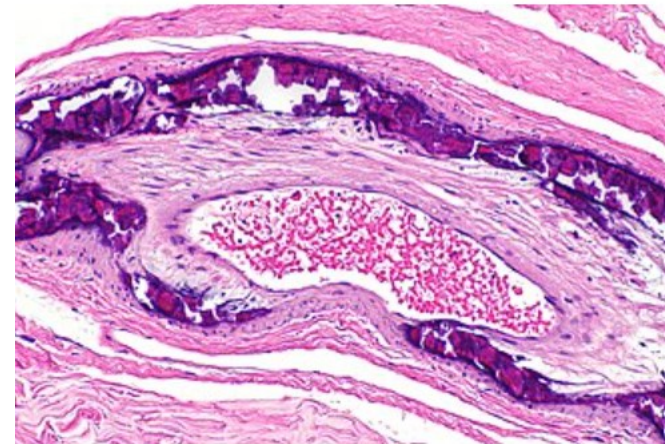
B. Mönckeberg Medial Calcific Sclerosis

1. Pathology:

- ❑ Calcification of the **tunica media** of small to medium-sized muscular arteries (e.g., radial, uterine).
 - The lumen is **not narrowed**.

2. Significance:

- ❑ Often incidental on X-ray, but can be associated with diabetes mellitus and chronic kidney disease.



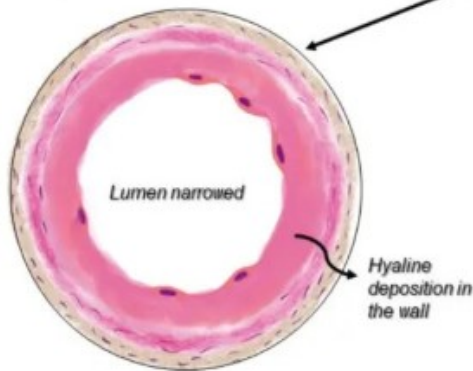
C. Arteriolosclerosis

Pathology: Affects small arteries and arterioles.
Two types:

- 1. Hyaline Arteriolosclerosis:** Homogeneous, pink hyaline thickening of the arteriolar wall.
 - Classic finding in **hypertension** and **diabetes mellitus**.
 - Causes luminal narrowing and end-organ damage (e.g., kidney, retina).
- 2. Hyperplastic Arteriolosclerosis:** "Onion-skin" concentric thickening of the wall.
 - Characteristic of severe (malignant) hypertension.
 - Can lead to fibrinoid necrosis.

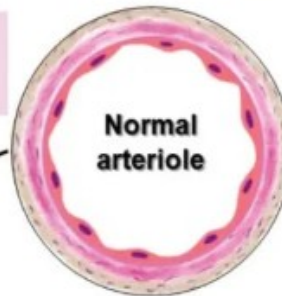
HYPERTENSIVE ARTERIOLOSCLEROSIS

Hyaline Arteriolosclerosis

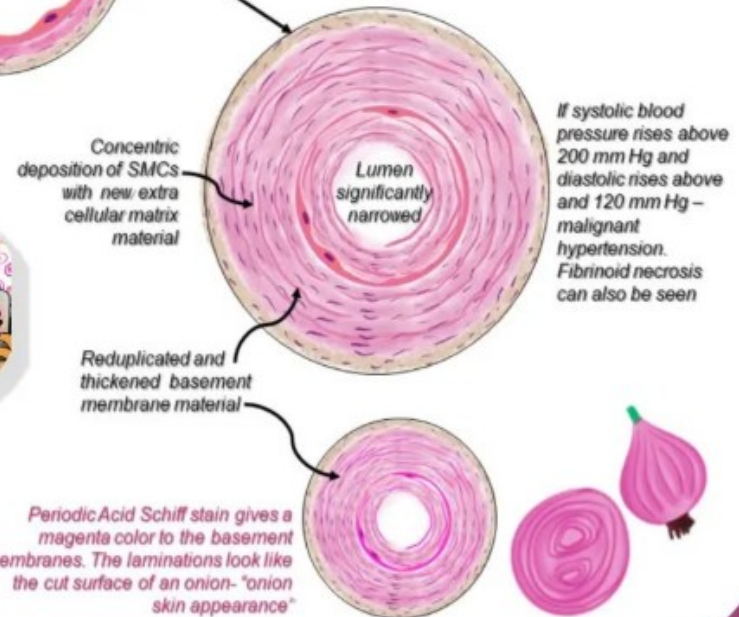


Injury to endothelial cells results in plasma protein leakage
+
SMCs make some new extra cellular matrix material
=
Wall thickens and lumen narrows

Normal arteriole



Hyperplastic Arteriolosclerosis



Vasculitis

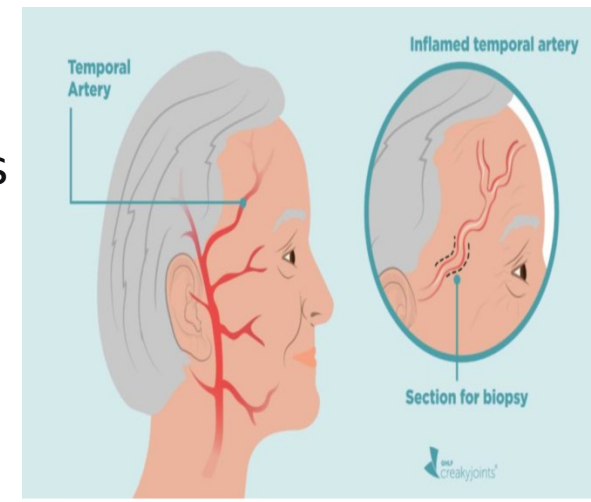
- **Inflammation** of Vessel Walls
- A heterogeneous group of diseases classified by the predominant **size** of the affected vessels.
- Pathogenesis often involves immune complexes
 - E.g., HBV, HCV, ANCA (Anti-Neutrophil Cytoplasmic Antibodies), or direct autoimmune attack.

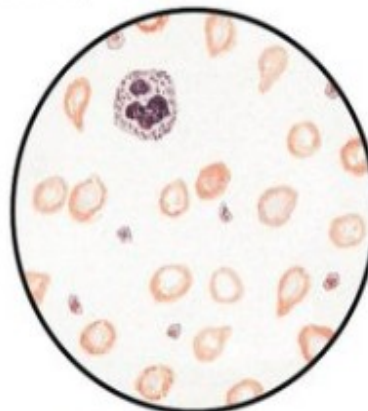
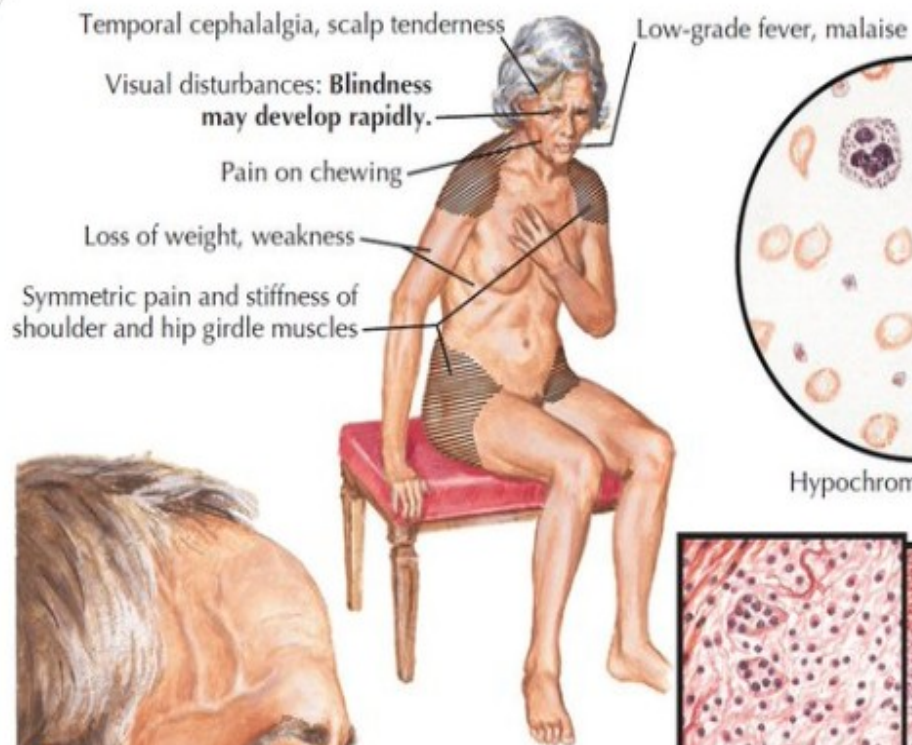
Vasculitis

Large-Vessel Vasculitis

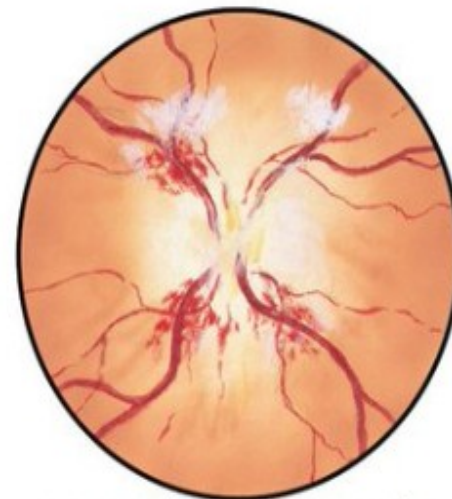
1. Giant Cell (Temporal) Arteritis:

- ❑ Granulomatous inflammation of cranial arteries.
 - Temporal, ophthalmic, posterior ciliary arteries
- ❑ The primary target is the **elastic lamina** of the artery.
- ❑ Activated **CD4+ T-cells** and **macrophages** are the central effector cells.
- ❑ **A classic histologic image of GCA**
 - **Transmural inflammation, giant cells, and a disrupted internal elastic lamina.**
- ❑ Risk of **blindness** if ophthalmic artery involved.
- ❑ Associated with polymyalgia rheumatica.

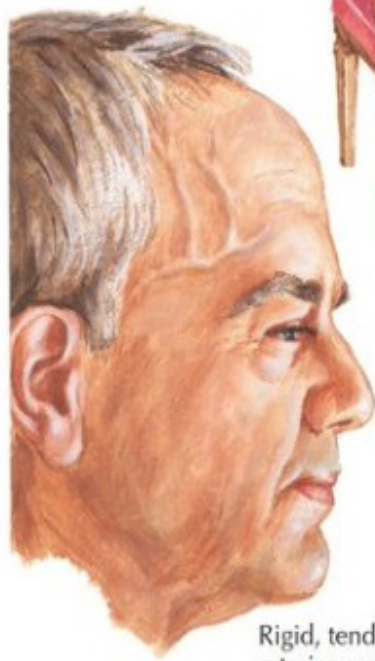




Hypochromic anemia

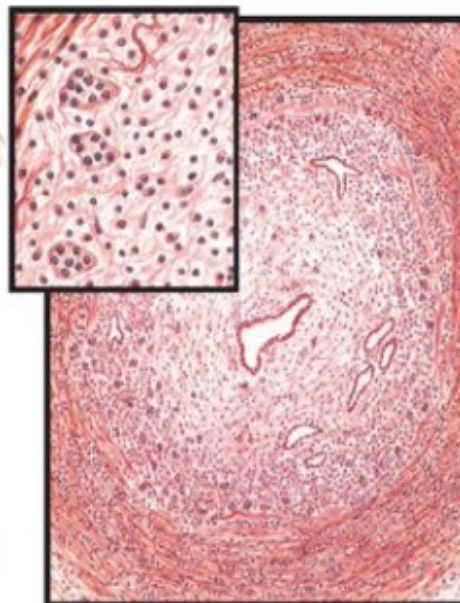


Anterior ischemic optic neuropathy



Rigid, tender, nonpulsating temporal arteries may be visible or palpable.

F. Netter M.D.



Elevated sedimentation rate

Biopsy specimen of superficial temporal artery: almost total obliteration of lumen with some recanalization. High-power insert shows infiltration with lymphocytes, plasma cells, and giant cells; fragmentation of elastica.

Large-Vessel Vasculitis

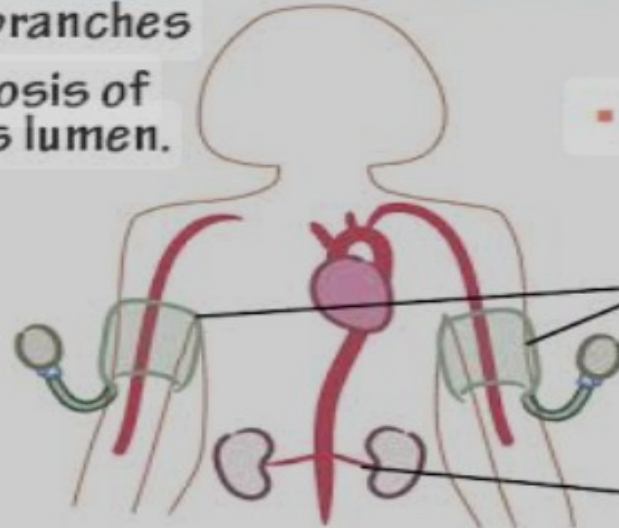
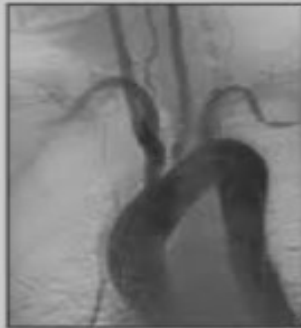
2. Takayasu Arteritis

- ❑ A chronic, granulomatous vasculitis of **large vessels**
 - begins at the vasa vasorum and progresses to devastating transmural fibrosis
- ❑ Primarily affects the **aorta and its main branches** ("pulseless disease")
- ❑ **Target:** The **media** of the large elastic arteries
- ❑ **Key Players:** Cytotoxic CD8+ and CD4+ T-cells, macrophages, and natural killer (NK) cells
- ❑ The inflammation is often **full-thickness** (panarteritis)
 - **granuloma formation** with giant cells and patchy necrosis of the media
 - **fibroblast proliferation** and scarring: **Tree-Bark appearance**
 - results in the characteristic triad of **stenosis, occlusion, and aneurysm formation** in the great vessels.
- ❑ Primarily in young women.

(aka, Pulseless disease)
Granulomatous Disease

✓ Aorta and its large branches

- Thickening and fibrosis of vessel wall narrows lumen.



- Ischemia produces:

“Pulseless disease”
Weak/absent pulse.

Different pressures
in upper extremities.
Claudication in limbs,
chest pain.

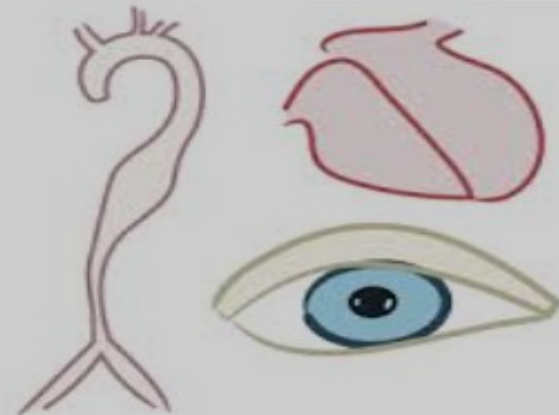
Poss. hypertension
(renal art. stenosis)

- Nonspecific symptoms assoc.
with inflammation:

Weakness, fatigue, fever,
weight loss, arthralgia.

✓ Poss. complications:

Aneurysm, aortic regurgitation,
retinopathy.



✓ Most common in women < 40 y. o.; esp. Asian ancestry.

Vasculitis

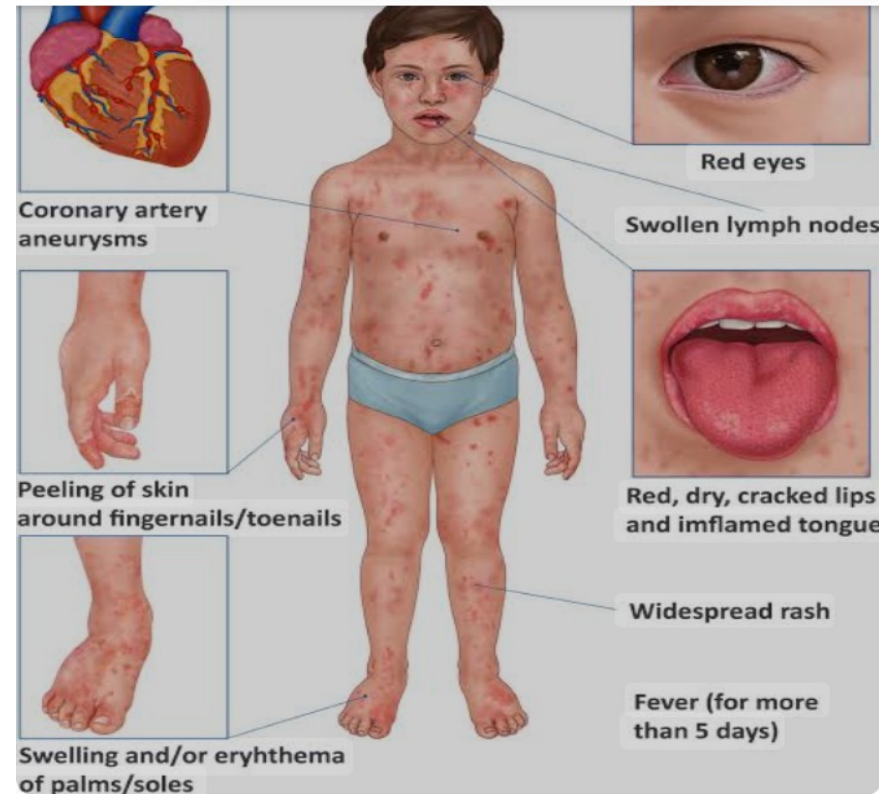
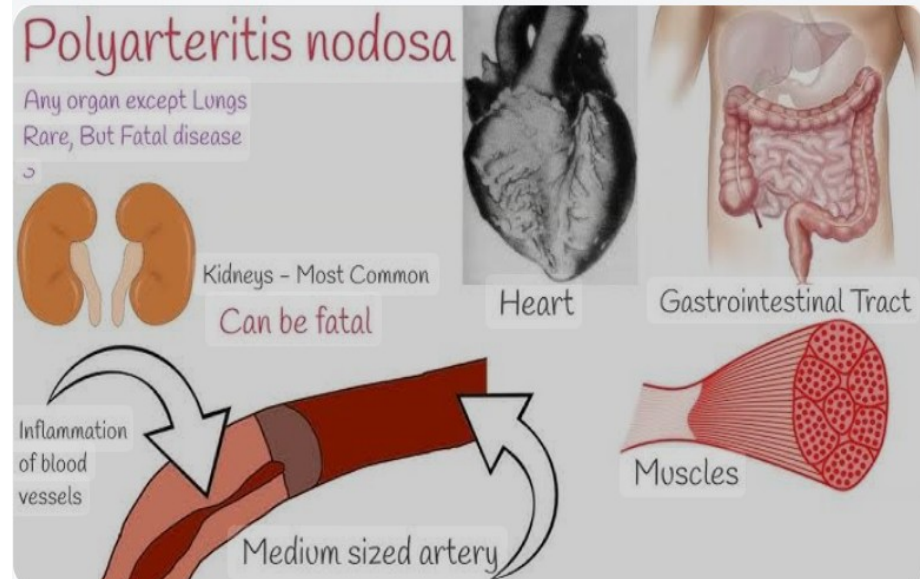
Medium-Vessel Vasculitis

1. Polyarteritis Nodosa (PAN)

- ❑ Necrotizing inflammation without granulomas,
- ❑ often involve renal, coronary, and GI vessels.
- ❑ Associated with **HBV**.

2. Kawasaki Disease

- ❑ Acute, necrotizing vasculitis of childhood
- ❑ affect coronary arteries, often causing aneurysms.



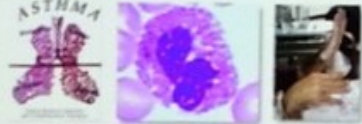
Vasculitis

Small-Vessel Vasculitis

- ❑ **ANCA-Associated:** Involve capillaries, venules, and arterioles.
 - ❖ Granulomatosis with polyangiitis (GPA)
 - ❖ Microscopic polyangiitis (MPA)
 - ❖ Eosinophilic granulomatosis with polyangiitis (EGPA).
- ❑ **Immune Complex-Mediated:**
 - ❖ IgA vasculitis (Henoch-Schönlein Purpura)
 - ❖ cryoglobulinemic vasculitis,
 - ❖ lupus vasculitis.

ANCA-associated vasculitis

Three genetically distinct syndromes with overlapping clinical features

	Clinical Phenotype	Histology	Seromarkers	Genetic Association
GPA		granulomatous inflammation + vasculitis	PR3-ANCA	HLA-DP SERPINA-1 PRTN-3
MPA		vasculitis	MPO-ANCA	HLA-DQ
EGPA		eosinophilia +/- vasculitis	MPO-ANCA (30 % of patients)	HLA-DQ (MPO-ANCA+) GPA33/IL5 (ANCA-)

GPA: Granulomatosis with Polyangiitis

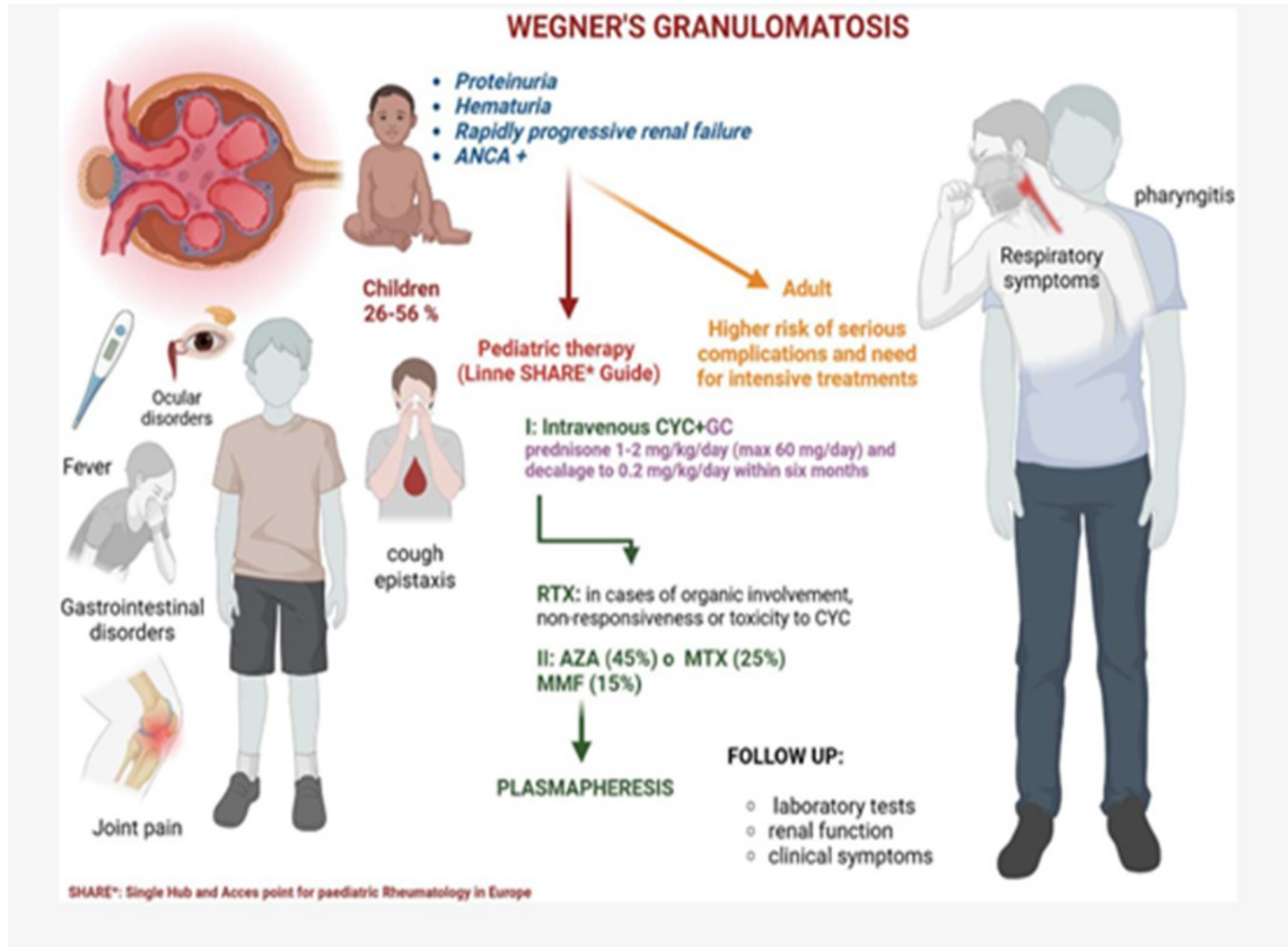
MPA: Microscopic Polyangiitis

GPA: Granulomatosis with Polyangiitis

Lyons PA et al. *New Engl J Med* 2012;367:214-23

Lyons PA et al. *Nature Communications* 2019; 10: 5120 ff

Granulomatosis with polyangiitis (GPA)



Microscopic Polyangiitis

Microscopic Polyangiitis (Small vessel polyarteritis)

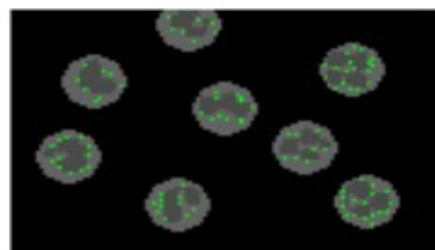
Easy to diagnose and treat -- if you think of it.

Smaller veins and arteries show patchy 3-layer inflammation

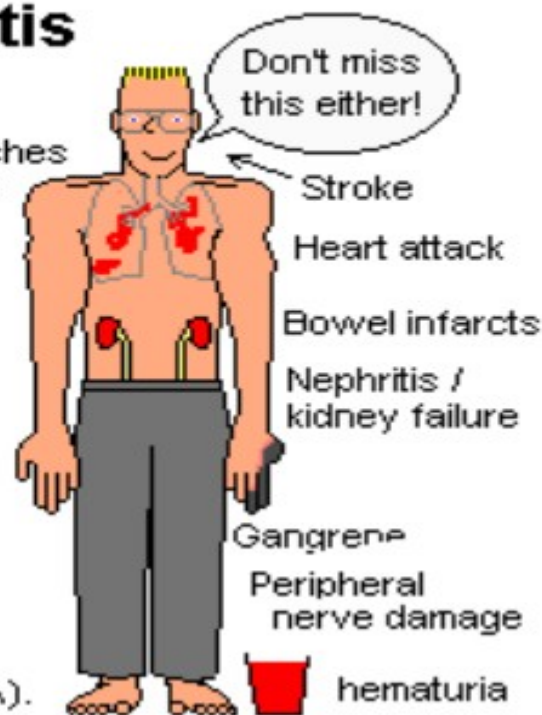


Vague aches and pains

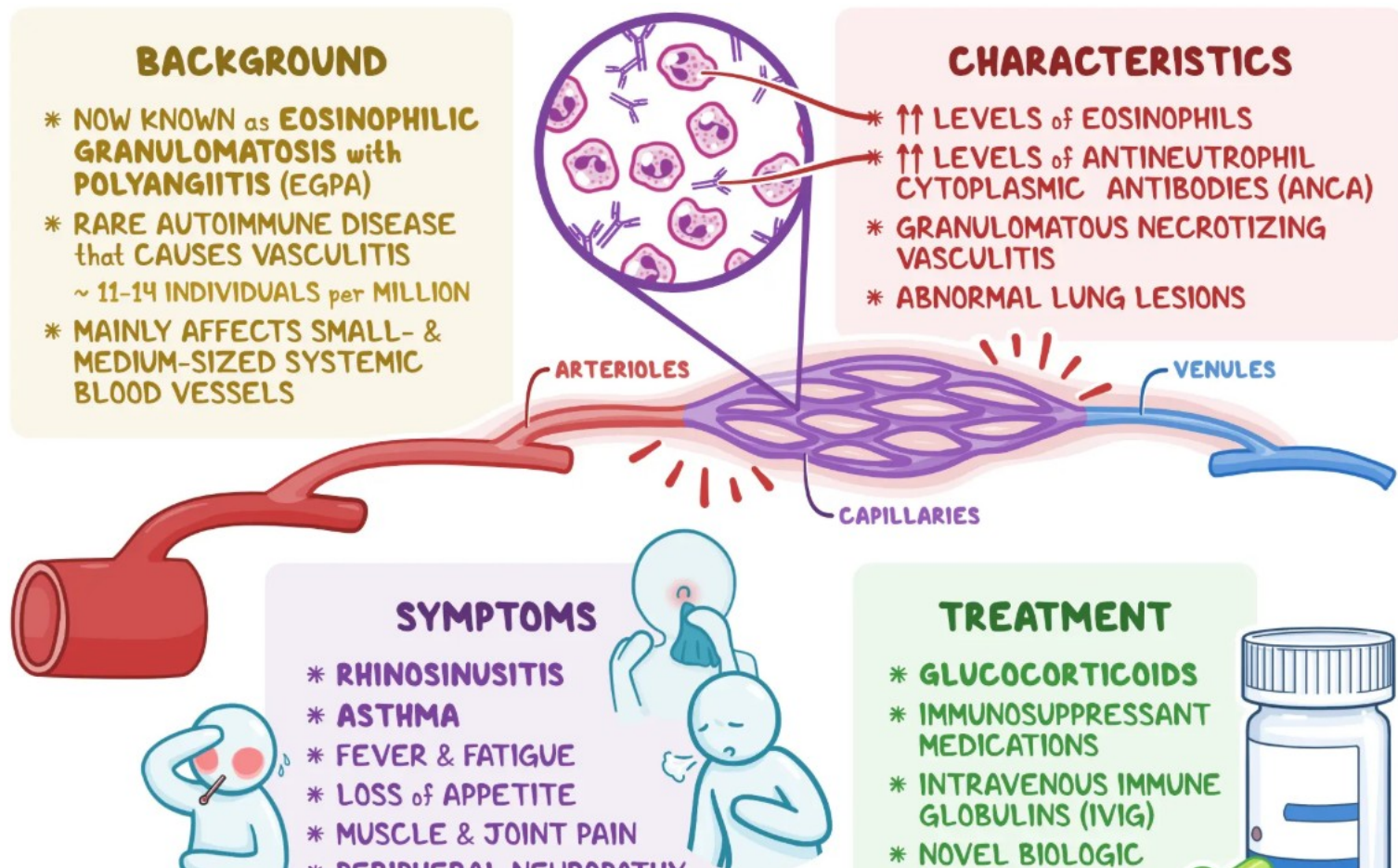
Hemoptysis & infiltrates



Positive anti-neutrophil cytoplasm test (p-ANCA).



Eosinophilic granulomatosis with polyangiitis(EGPA).



IgA vasculitis (Henoch-Schönlein Purpura)



Aneurysms and Dissections

Aneurysm: Abnormal, permanent dilation of a vessel by $\geq 50\%$ of its normal diameter.

- **True Aneurysm:**

- ☐ Involves all three layers of the vessel wall
 - ❖ e.g., atherosclerotic abdominal aortic aneurysm

- **False Aneurysm (Pseudoaneurysm):**

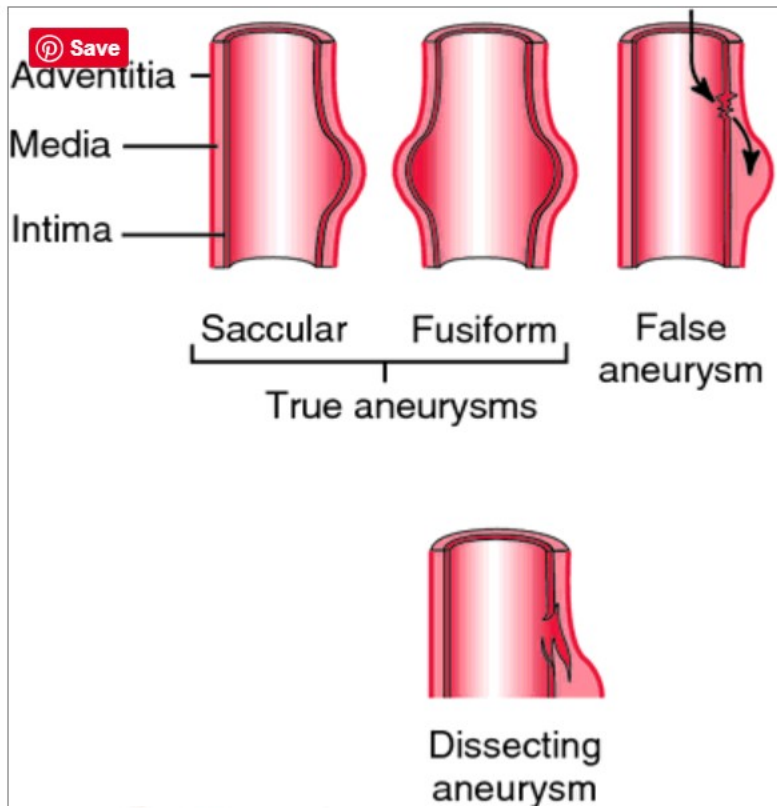
- ☐ A breach in the wall contained by a clot or adventitia.
- ☐ **Etiology:**
 - ❖ Atherosclerosis (most common),
 - ❖ hypertension, genetic defects (e.g., Marfan syndrome - defective fibrillin),
 - ❖ infection ("mycotic" aneurysm), trauma.

Aortic Dissection:

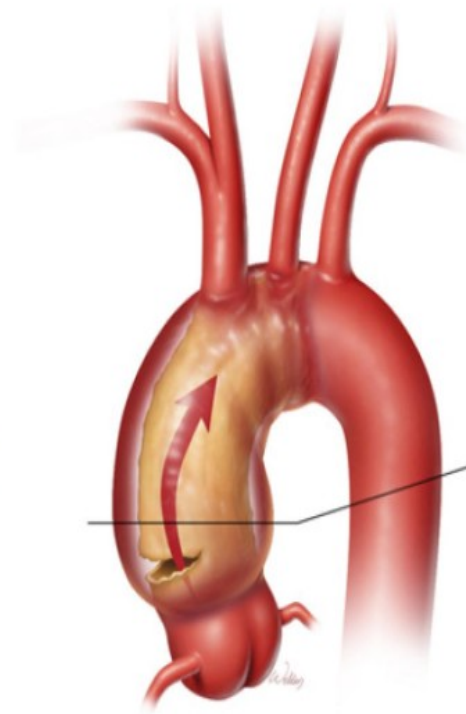
- ❑ A tear in the intima allows blood to enter the wall of the aorta, dissecting through the media and creating a false lumen.
- ❑ A catastrophic event.

- **Risk Factors:**

- Hypertension (most common),
- connective tissue disorders (Marfan, Ehlers-Danlos),
- bicuspid aortic valve.

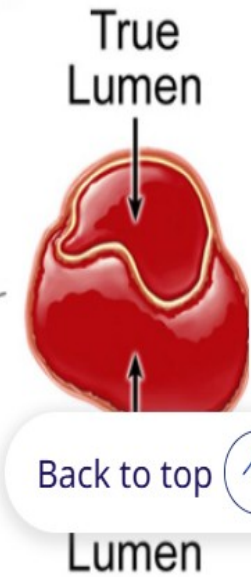


Aneurysm



Dissection

[Credit](#)



Vascular Tumors

- **Benign:** Hemangioma (capillary, cavernous), Lymphangioma.
- **Intermediate (Locally Aggressive):** Kaposi Sarcoma (caused by Human Herpesvirus 8).
- **Malignant:** Angiosarcoma (highly aggressive, often on skin/scalp of elderly).

Vascular Tumors

- Vascular tumors (angiomas and angiosarcomas) are a heterogeneous group of neoplasms arising from
 - blood vessels (hemangio-) or
 - lymphatic vessels (lymphangio-).
- Their behavior ranges from benign and self-limiting to highly aggressive and metastatic.
- Diagnosis relies heavily on histopathological examination, often supplemented by immunohistochemical markers like **CD31**, **CD34**, **ERG**, and **D2-40** (for lymphatic differentiation).

Benign Vascular Tumors

- These are common, often congenital, and have an excellent prognosis.

1. Infantile Hemangioma

- **Clinical:** The most common tumor of infancy.
- Presents weeks after birth as a rapidly growing red ("strawberry") lesion.
- Undergoes a **proliferative phase** (first year), followed by a slow **involution phase** over years.
- Most regress completely without treatment.
- **Pathology:**
 - **Proliferative Phase:** Well-circumscribed, unencapsulated.
 - Dense proliferation of plump, mitotically active endothelial cells forming small, capillary-sized vessels.
 - Lumina may be inconspicuous.
 - **Involuting Phase:** Endothelial cells flatten, apoptosis increases, and vessels dilate. Progressive fibrosis and fatty deposition replace the cellular areas.
- **Key Feature: GLUT1 positive** (immunohistochemical marker) throughout all phases, distinguishing it from other vascular malformations and tumors.

Benign Vascular Tumors

Congenital Hemangioma

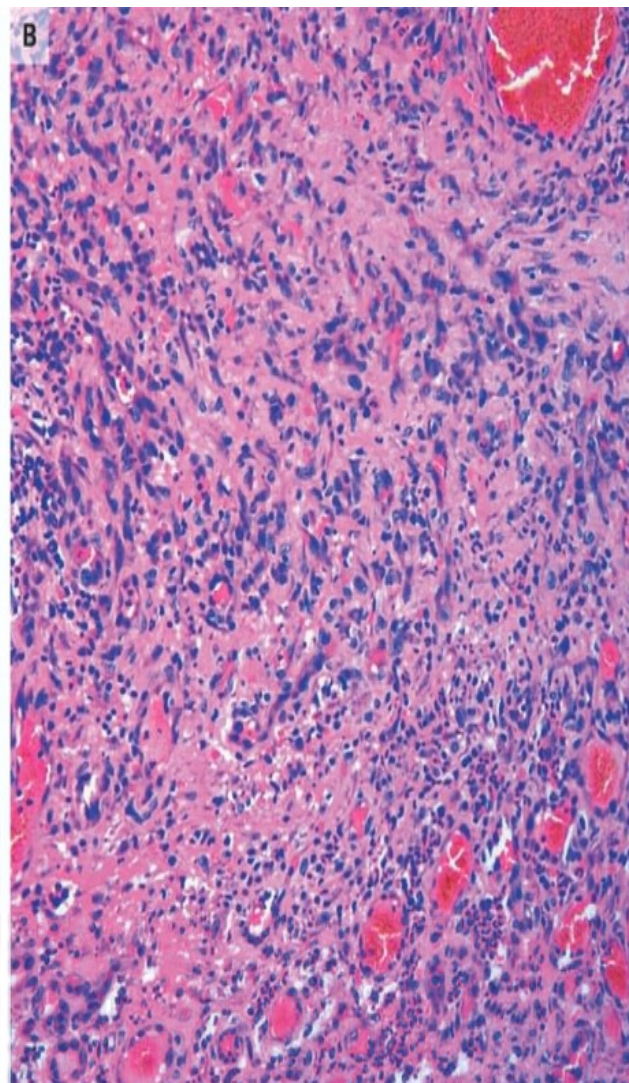
- **Clinical:**
 - ❑ Present fully grown at birth
 - ❖ RICH: Rapidly Involuting Congenital Hemangioma) or
 - ❖ do not involute (NICH: Non-Involuting Congenital Hemangioma).
- **Pathology:** Histology similar to infantile hemangioma but **GLUT1 negative**.

Cavernous Hemangioma

- **Clinical:**
 - ❑ Often deeper than capillary hemangiomas, found in liver, brain, and soft tissue.
 - ❑ Does not involute spontaneously.
- **Pathology:**
 - ❑ Large, dilated, blood-filled vascular channels ("cavernous") lined by flattened, benign endothelial cells.
 - ❑ The vessels are often separated by fibrous stroma. May contain phleboliths (calcified thrombi).
- **Key Feature:**
 - ❑ Can be part of **von Hippel-Lindau syndrome** (central nervous system lesions) or other syndromes.

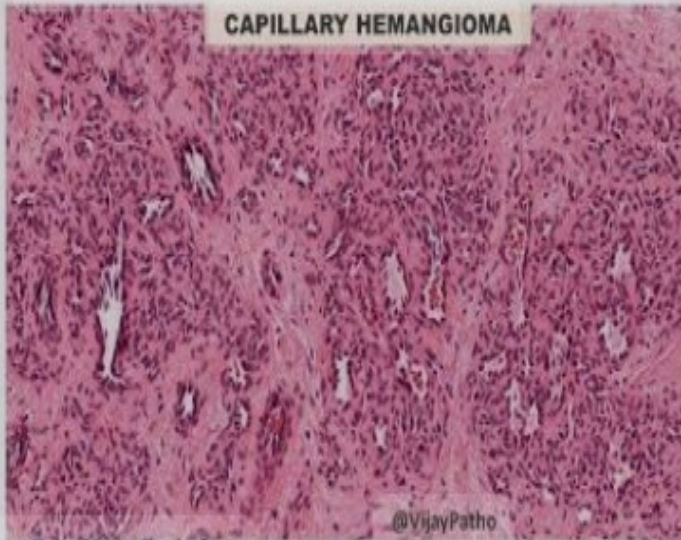
Capillary Hemangioma (Pyogenic Granuloma/Lobular Capillary Hemangioma)

- **Clinical:**
 - ❑ A reactive, often rapidly growing polypoid lesion on skin, gingiva, or in pregnancy ("pregnancy tumor").
 - ❑ Frequently ulcerates and bleeds.
- **Pathology:**
 - ❑ Exophytic growth with a **lobular architecture**.
 - ❑ Each lobule contains proliferating capillaries with plump endothelial cells, often with a prominent feeder vessel.
 - ❑ Surface ulceration and acute inflammation are common.
- **Key Feature:**
 - ❑ Despite the name, it is not infectious ("pyogenic") but rather reactive/benign.



CAPILLARY

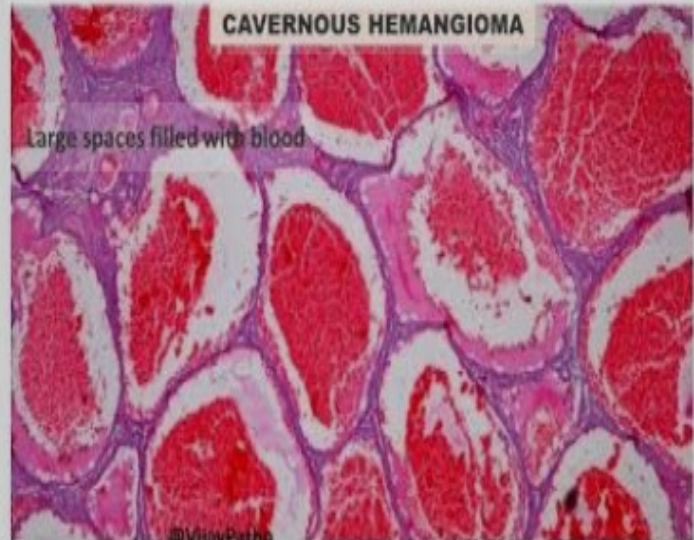
CAPILLARY HEMANGIOMA



@VijayPatho

CAVERNOUS

CAVERNOUS HEMANGIOMA

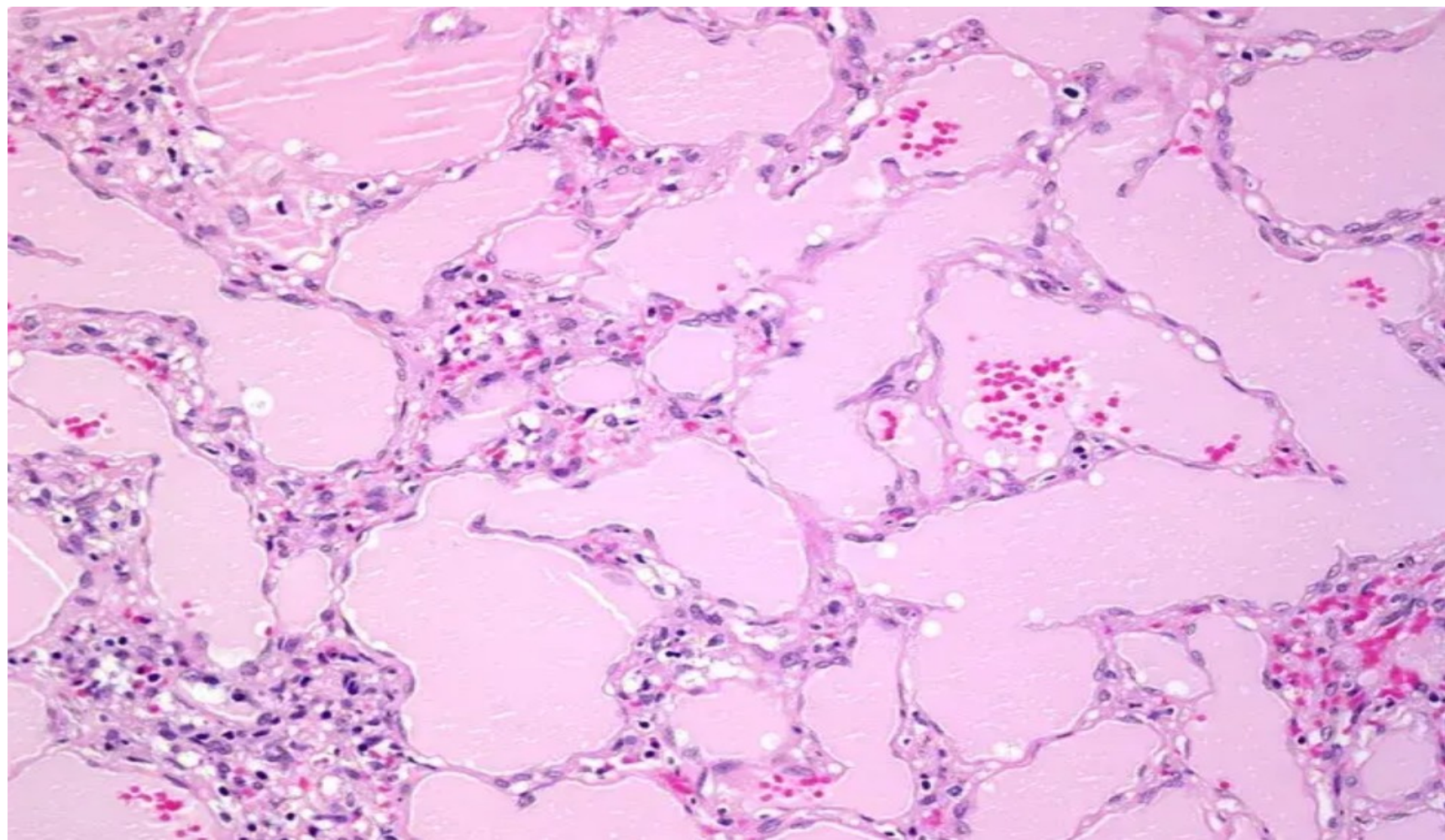


Large spaces filled with blood

@VijayPatho

Lymphangioma

- **Clinical:**
 - ❑ Benign lymphatic channel malformations.
 - ❑ **Cystic hygroma** is a large type typically in the neck/axilla of children.
 - ❑ **Simple lymphangioma** is smaller and superficial.
- **Pathology:**
 - ❑ Dilated, thin-walled lymphatic channels lined by flat endothelial cells.
 - ❑ Channels contain proteinaceous lymph fluid, not blood.
 - ❑ The stroma often contains lymphoid aggregates.
- **IHC:** Positive for lymphatic markers **D2-40 (podoplanin)** and PROX1.



Intermediate-Grade (Locally Aggressive) Vascular Tumors

- These tumors rarely metastasize but are locally destructive and have a high rate of recurrence.

1. Kaposi Sarcoma (KS)

Etiology:

- ☐ Caused by **Human Herpesvirus 8 (HHV-8)**.
- ☐ Associated with immunosuppression (HIV/AIDS, post-transplant).

Clinical Variants:

- ☐ Classic (elderly Mediterranean men)
- ☐ Endemic (African)
- ☐ Iatrogenic (immunosuppressed)
- ☐ AIDS-associated (aggressive, often visceral)

TYPES



Classic Kaposi's Sarcoma



Endemic Kaposi's Sarcoma



Immunosuppression Therapy-Related Kaposi's Sarcoma



Epidemic Kaposi's Sarcoma

FEATURES

- Common in older adults (especially men)
- Slow-growing
- Lesions in the legs

- Common in young adults (more in men) in Africa
- Often aggressive

- Occurs in organ transplantation patients.
- Mostly affects the skin

- Seen in AIDS patients
- Affects multiple areas of the body

Intermediate-Grade (Locally Aggressive) Vascular Tumors

Pathology:

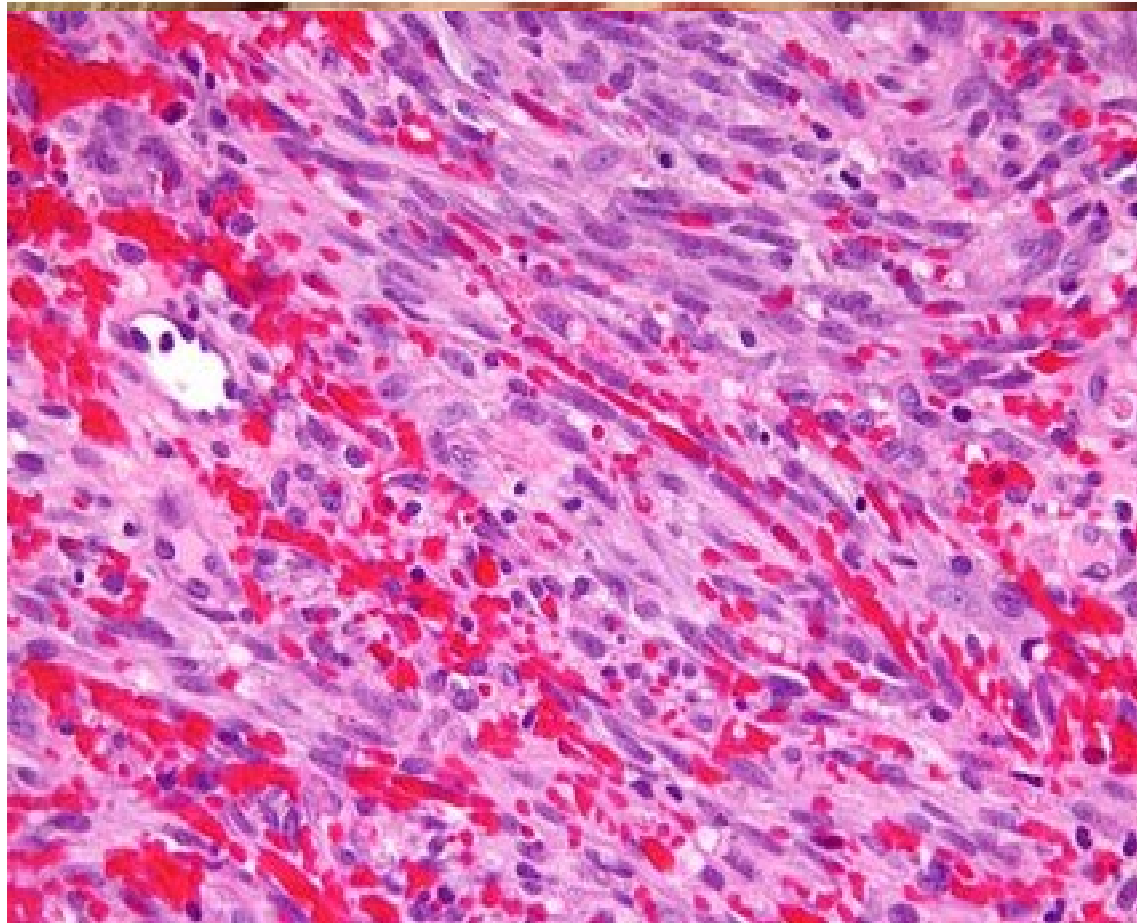
- **Early Patch Stage:**

- ☐ Irregular, jagged vascular channels dissecting dermal collagen, with mild lymphoplasmacytic infiltrate.

- **Plaque/Nodular Stage:**

- ☐ More cellular, forming discrete nodules of spindle-shaped cells.
- ☐ Slit-like vascular spaces containing extravasated red blood cells are characteristic.
- ☐ **Hyaline globules** (degenerated erythrocytes) are often seen.

- **IHC:** The spindle cells are positive for vascular markers (CD31/34) and, most importantly, **HHV-8 Latent Nuclear Antigen (LANA-1)**.



Papillary Endothelial Hyperplasia (Masson Tumor)

Clinical:

- ☐ A reactive, organizing thrombus that can mimic angiosarcoma.
- ☐ Presents as a small, tender mass, often in a vein.

Pathology:

- ☐ Papillary fronds with hyaline cores covered by a single layer of bland endothelial cells.
- ☐ The fronds are essentially organizing thrombus.
- ☐ Lack of cytologic atypia, necrosis, and mitotic activity distinguishes it from angiosarcoma.

IntraVascular papillary endothelial hyperplasia (masson's tumour)

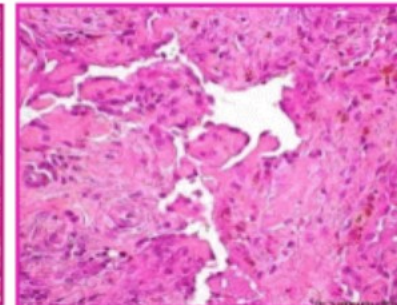
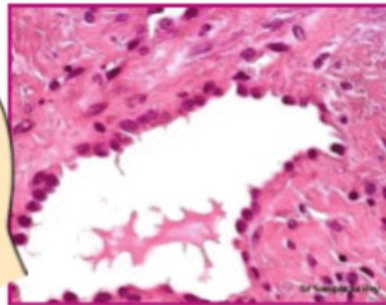
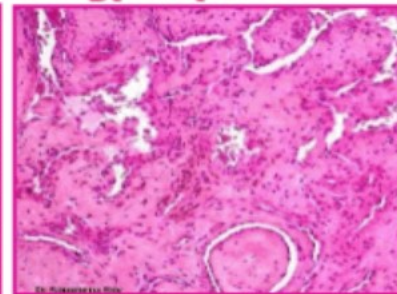
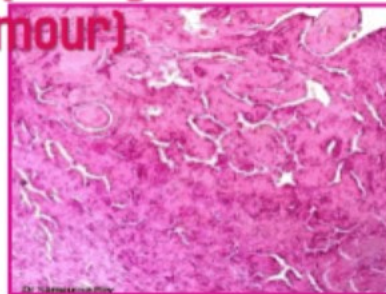


First described by
Pierre Masson in
1923

Presents as a
small, firm, blue
or purple nodule



Occur in any blood
vessel in the body
Common sites -
Fingers, head & neck
and trunk



#roypath

histopathology-india.net

Malignant Vascular Tumors

- These are rare but highly aggressive sarcomas with a poor prognosis.

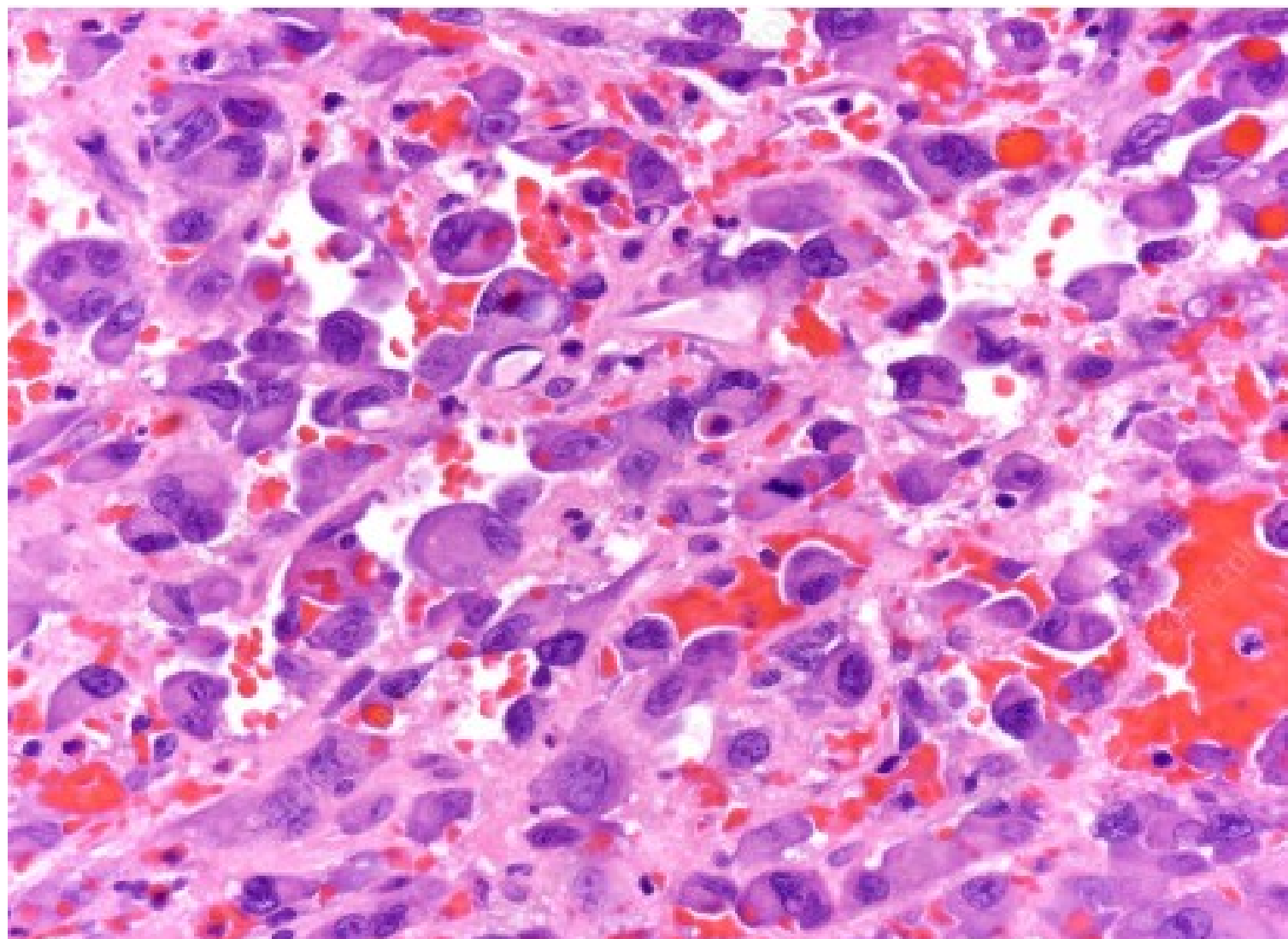
Angiosarcoma

- **Clinical:** Arises in **four main settings**:
 - **Skin and Soft Tissue of Elderly:** Head and scalp most common site.
 - **Chronic Lymphedema: Stewart-Treves syndrome** (angiosarcoma in a chronically lymphedematous arm post-mastectomy).
 - **Post-Radiation Therapy:** Develops in a previous radiation field (e.g., for breast cancer), often after a long latency.
 - **Parenchymal Organs:** e.g., Liver (associated with vinyl chloride, arsenic, Thorotrast exposure), breast, deep soft tissue.

Angiosarcoma

Pathology:

- **Gross:** Poorly circumscribed, hemorrhagic, spongy mass ("blood-filled cyst").
- **Microscopic: Extreme histologic variability**, from well-differentiated to poorly differentiated.
 - **Well-differentiated:**
 - Irregular, anastomosing vascular channels that **dissect through collagen and adipose tissue**.
 - Endothelial cells are hyperchromatic and multilayered.
 - **Poorly differentiated:**
 - Solid sheets of malignant spindle or epithelioid cells with marked pleomorphism, high mitotic rate, and necrosis.
 - Vasoformation may be subtle and require IHC confirmation.
- A common and ominous feature is **intratumoral hemorrhage** due to the fragility of the malignant vessels.
- **IHC:** Positive for **CD31** (most sensitive and specific), CD34, ERG, FLI-1.
- **Prognosis:** Generally poor due to high local recurrence and early hematogenous metastasis (lungs, liver, bone). Scalp tumors have a particularly dismal prognosis.



Epithelioid Hemangioendothelioma (EHE) - Intermediate to Malignant

Clinical:

- ❑ A vascular tumor of intermediate malignancy that can arise in soft tissue, bone, liver, lung, and other sites.
- ❑ Behavior is unpredictable; some behave indolently, while others metastasize.

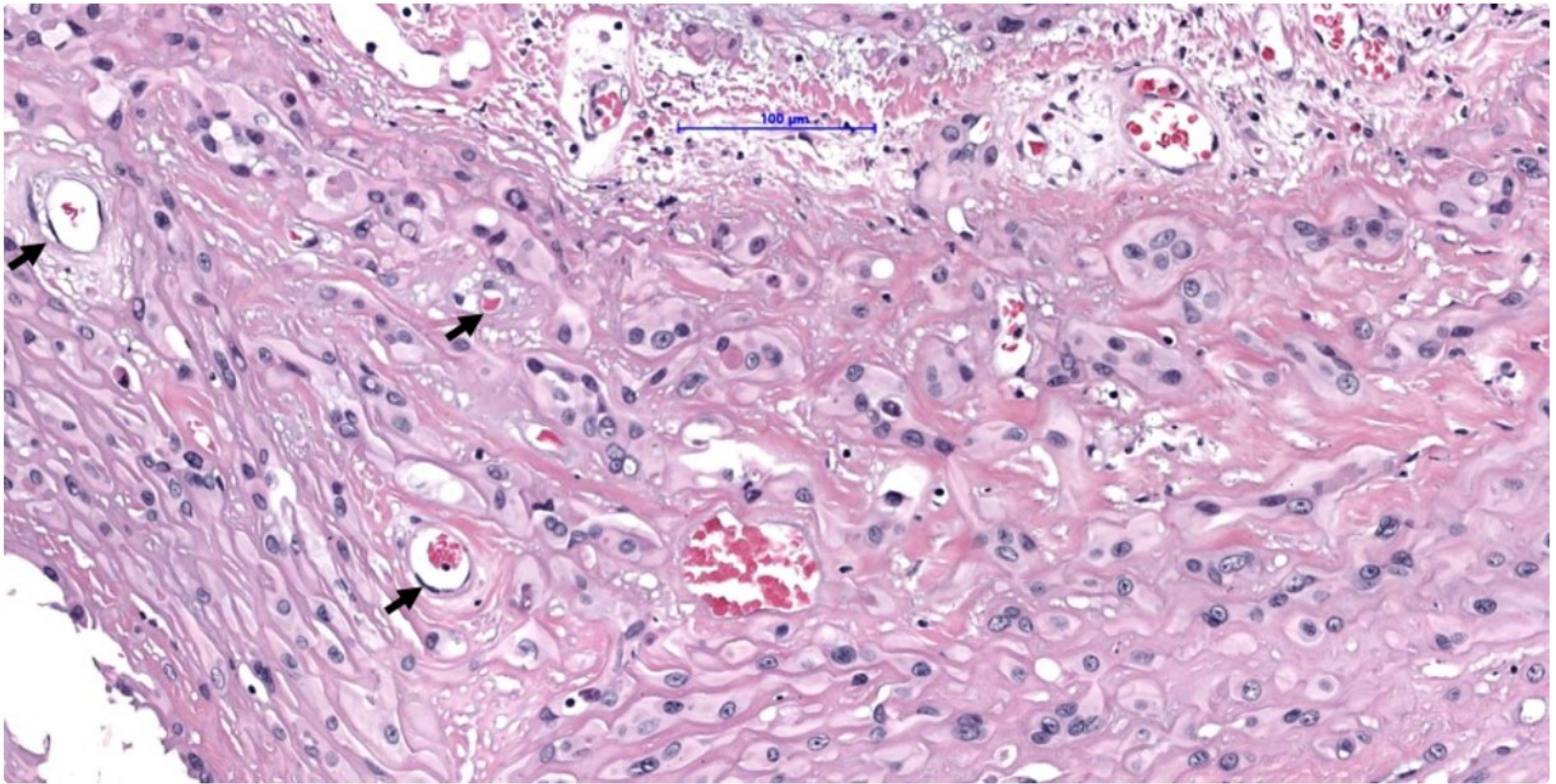
Pathology:

- ❑ Cords, nests, or short strands of **epithelioid endothelial cells** embedded in a distinctive **myxohyaline stroma**.
- ❑ The cells often contain intracytoplasmic lumina (**vacuoles**) that may contain red blood cells ("blister cells").

Genetic hallmark: WWTR1::CAMTA1 gene fusion (90% of cases) or a rare **YAP1::TFE3 fusion**.

IHC: Positive for vascular markers (CD31/34) and **CAMTA1** (for WWTR1-rearranged tumors).

Epithelioid Hemangioendothelioma (EHE)



Venous Pathology

Varicose Veins:

- ❑ Abnormally dilated, tortuous veins due to increased intraluminal pressure and loss of vessel wall support.
- ❑ Common in superficial leg veins (saphenous).

Phlebothrombosis & Thrombophlebitis:

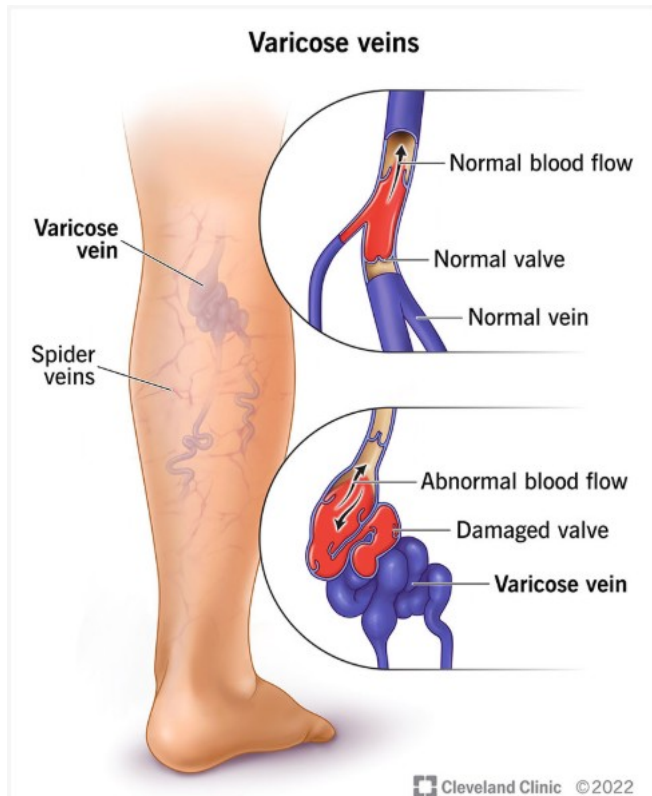
- ❑ Venous thrombosis.
- ❑ **Deep Vein Thrombosis (DVT)** in large deep veins (e.g., femoral, iliac) is dangerous due to risk of **pulmonary thromboembolism**.

Vascular Dysfunction

Raynaud Phenomenon:

- ❑ Exaggerated vasoconstriction of digital arteries and skin arterioles in response to cold or emotion, causing color changes (white → blue → red).

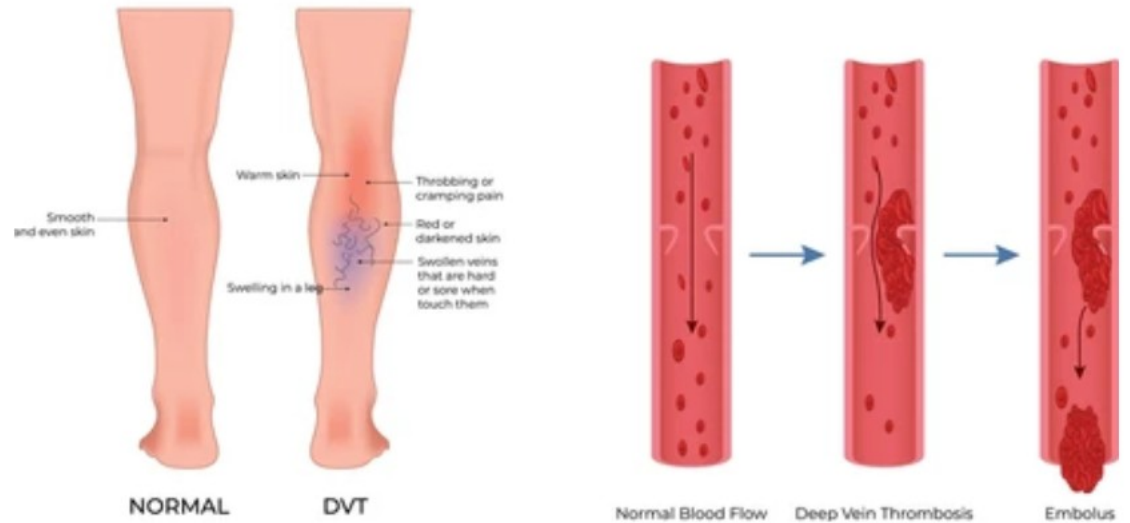
Varicose Veins



Thrombophlebitis



Deep Vein Thrombosis (DTV)





Lymphatic Vessel Pathology

I. Normal Structure & Function

- Lymphatics are thin-walled, endothelial-lined channels that drain excess interstitial fluid (lymph), proteins, and cells (immune surveillance) from tissues back to the bloodstream.
- They lack a substantial muscular layer and have overlapping endothelial junctions that act as one-way valves.

II. Major Pathologic Processes

A. Lymphoedema

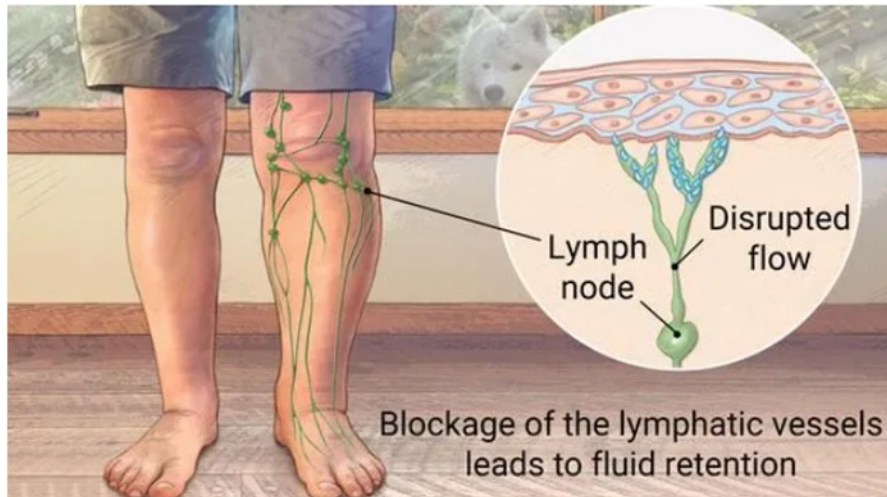
- Accumulation of protein-rich interstitial fluid due to impaired lymphatic drainage.

Causes:

- ☐ **Primary (Congenital):** Rare, due to developmental abnormalities (e.g., Milroy disease).
- ☐ **Secondary (Acquired):** Much more common.
 - **Obstruction:** Tumor compression, surgical resection of lymph nodes (e.g., in breast cancer axillary node dissection), **filariasis** (parasitic infection by *Wuchereria bancrofti*, a leading **global cause**).
 - **Inflammation:** Scarring from radiation therapy or recurrent infection.

Pathology:

- ☐ Chronic lymphoedema leads to stagnant lymph, impaired immunity, and dense dermal fibrosis.
- ☐ The skin becomes thick and tough ("peau d'orange" appearance).
- ☐ **Elephantiasis** is an extreme form with massive enlargement and fibrosis.



Leg lymphedema

Lymphangitis

- Inflammation of lymphatic channels, usually due to acute bacterial infection (commonly *Streptococcus pyogenes*).

Pathology:

- ☐ Bacteria drain from an infection site, causing painful, erythematous streaks under the skin (extending from the source to regional lymph nodes).
- ☐ The lymph nodes themselves become enlarged and painful (lymphadenitis).

Tumors & Malformations

Lymphangioma:

- ❑ Benign congenital malformation of lymphatic channels.
- ❑ Common in the neck (**cystic hygroma**).

Lymphangiosarcoma:

- ❑ A rare, highly aggressive malignancy that can arise in the setting of **chronic lymphoedema** (e.g., in the arm after radical mastectomy, known as Stewart-Treves syndrome).

